

# CRD: Assumptions & Diagnostics

## Checking Assumptions and Diagnosing Problems

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### Learning Objectives

By the end of this lecture, you should be able to:

- Create and interpret diagnostic plots: Q-Q plot, residual vs fitted, scale-location
- Conduct formal tests for assumptions: Shapiro-Wilk test, Levene's test

**Timing:** ~3 minutes

**Key message:** ANOVA results are only valid if assumptions hold. Today we learn to check them and fix violations.

**Connection to last lecture:** “Wednesday we learned ANOVA gives us p-values. Today we learn when to trust those p-values.”

## ANOVA Assumptions Explained

### Assumption 1: Normality

**What it means:**

- Errors (residuals) follow a normal distribution
- Not the raw data, but the **residuals** from the model

**Why it matters:**

- F-test relies on normal distribution theory
- Violated → p-values unreliable

**Good news:**

- ANOVA is **robust** to mild non-normality with balanced designs
- Central Limit Theorem helps with larger sample sizes ( $n > 30$  per group)

**Residual** = Observed - Predicted

$$e_{ij} = y_{ij} - \hat{y}_{ij}$$

where  $\hat{y}_{ij}$  is the treatment mean for group  $i$ .

**Timing:** ~4 minutes

**Common misconception:** “The data needs to be normal.” Correction: “The ERRORS need to be normal, which is different from data normality.”

## Assumption 2: Homogeneity of Variance

**What it means:**

- All treatment groups have the same error variance  $\sigma^2$
- Also called **homoscedasticity**

**Why it matters:**

- F-test assumes equal variance
- Violated → Some groups have noisier data, biasing comparisons

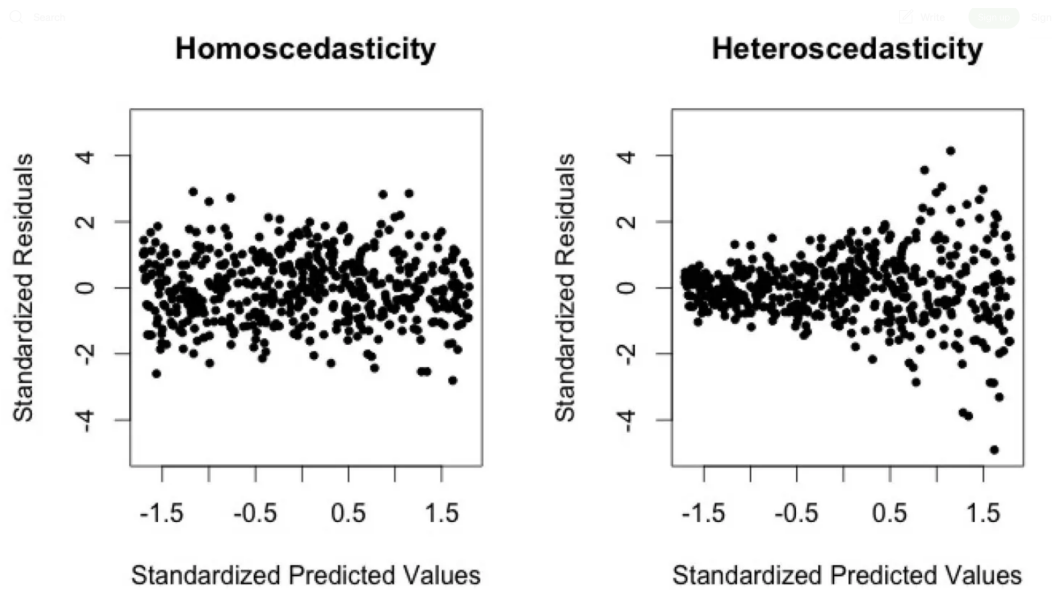


Figure 1: Homoscedasticity vs Heteroscedasticity

**Timing:** ~4 minutes

**Visual aid:** Draw two distributions with same mean but different spreads to illustrate heteroscedasticity.

### Assumption 3: Independence

**What it means:**

- Each observation is independent—no correlation between errors
- Randomization ensures independence

**Common violations in agriculture:**

- **Spatial correlation:** Adjacent plots influence each other (water runoff, pest movement)
- **Temporal correlation:** Repeated measures on same unit over time
- **Cluster effects:** Animals in same pen more similar than random

**How to prevent:**

- Proper randomization
- Physical separation of units
- Blocking to account for spatial patterns
- Use mixed models for repeated measures (later in course)

**Timing:** ~4 minutes

**Reassurance:** “Independence is mostly ensured by good experimental design. If you randomized properly, you’re usually OK.”

## Checking Normality

### The Q-Q Plot

#### Quantile-Quantile (Q-Q) Plot:

- Compares sample quantiles to theoretical normal quantiles
- **Good:** Points fall on diagonal line
- **Bad:** Points deviate systematically

#### Interpretation guide:

- **Light tails:** Flatter than line at ends
- **Heavy tails:** Steeper than line at ends
- **Skewness:** Curve away from line

**Timing:** ~4 minutes

**Teaching tip:** “Trust your eyes first—if the plot looks good, you’re probably fine. Formal tests are secondary.”

### Q-Q Plot Examples: Good vs Bad

**Timing:** ~3 minutes

**Interpretation:** Left plot shows right skew (common in count data, yield data). Right plot shows outliers. Both violate normality.

## Shapiro-Wilk Test for Normality

#### Formal hypothesis test:

- $H_0$ : Residuals are normally distributed
- $H_a$ : Residuals are not normally distributed

Good: Residuals are Normal

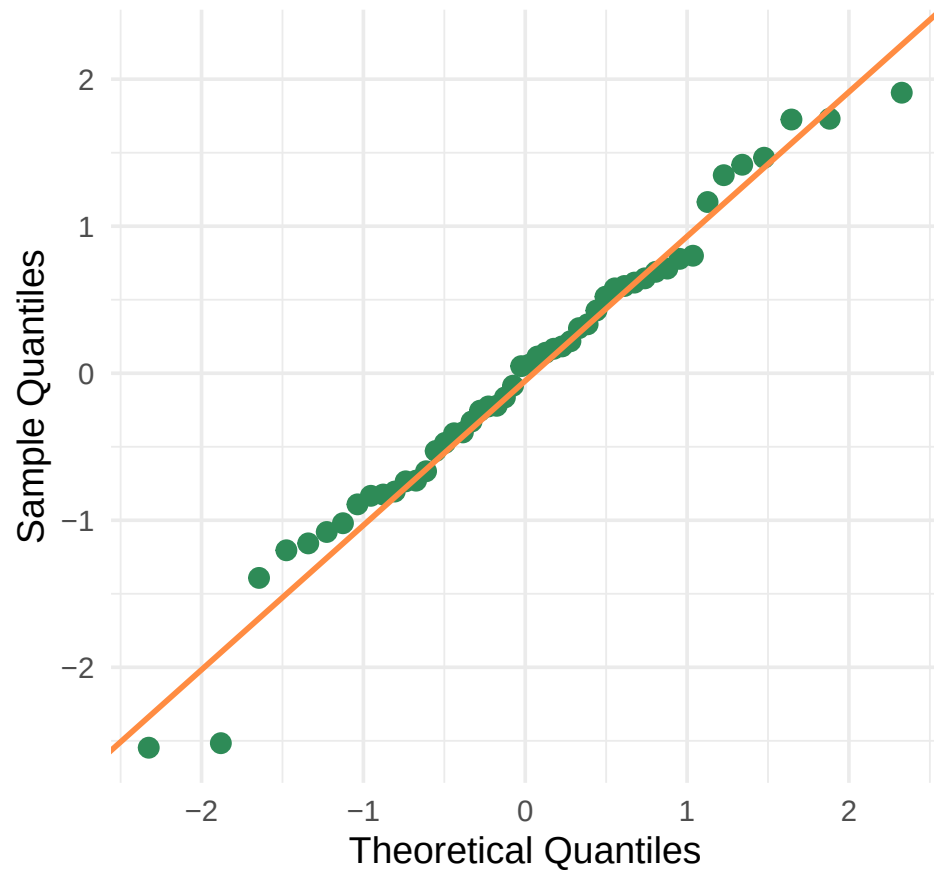


Figure 2: Good Q-Q plot: Points follow the line closely

## Bad: Right Skew

Points curve above line

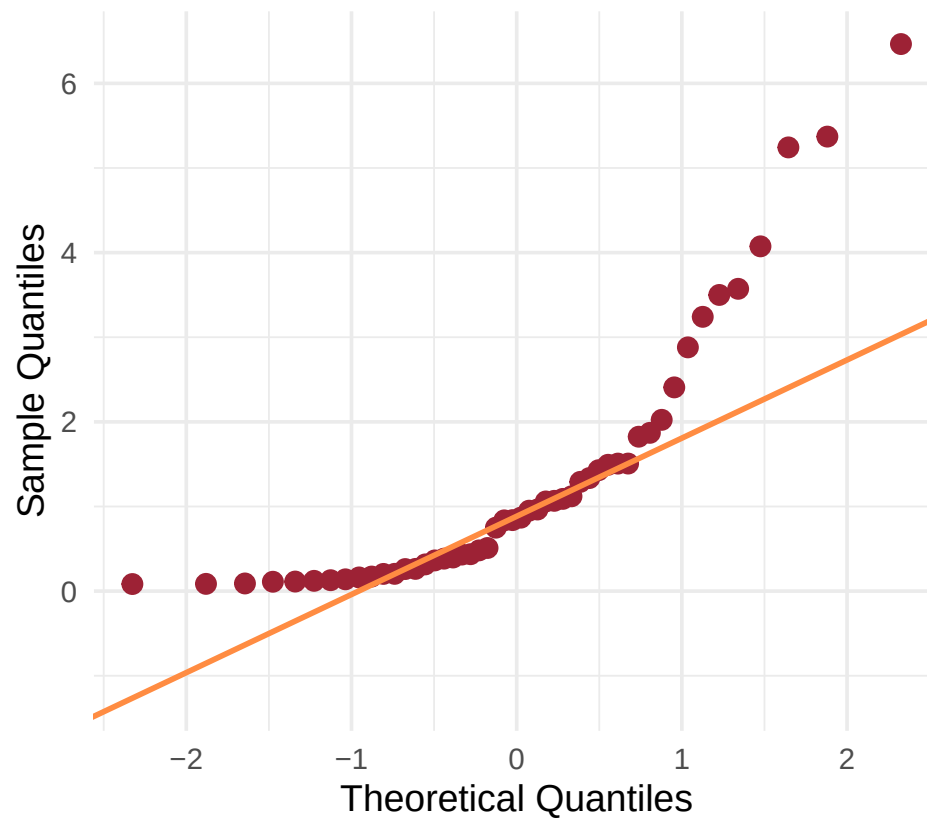


Figure 3: Bad: Right-skewed residuals (transformation needed)

## Bad: Heavy Tails

Points deviate at extremes

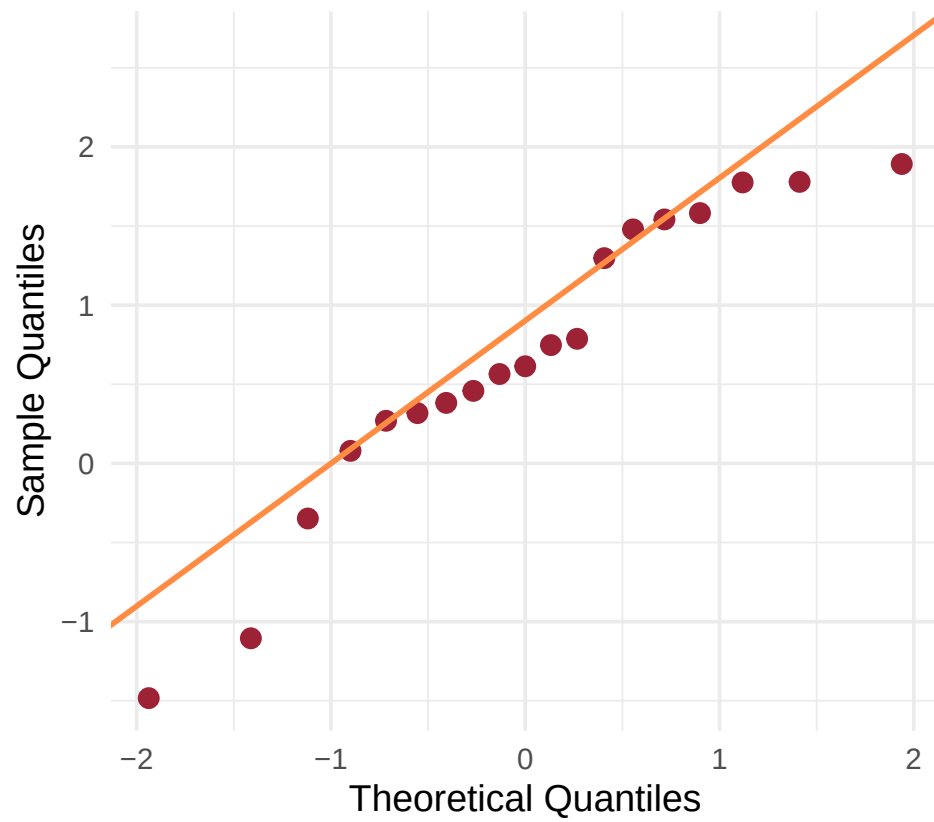


Figure 4: Bad: Heavy-tailed distribution (outliers present)

```

# Reload data from Lecture 3
set.seed(2026)
crd_data <- data.frame(
  treatment = rep(c("N0", "N50", "N100", "N150"), each = 6),
  yield = c(
    rnorm(6, 2400, 150), rnorm(6, 2650, 150),
    rnorm(6, 2900, 150), rnorm(6, 2950, 150)
  )
)

# Fit model
model_crd <- lm(yield ~ treatment, data = crd_data)

# Extract residuals
residuals <- resid(model_crd)

# Shapiro-Wilk test
shapiro.test(residuals)

```

```

#>
#>  Shapiro-Wilk normality test
#>
#> data:  residuals
#> W = 0.94396, p-value = 0.1998

```

### Interpretation:

- $p = 0.1998 > 0.05 \rightarrow$  Fail to reject  $H_0$
- No evidence of non-normality
- Safe to proceed with ANOVA

**Timing:** ~4 minutes

**Caution:** “Shapiro-Wilk is sensitive to sample size. Large samples can show ‘significant’ non-normality even when effect is trivial. Always check Q-Q plot too.”

## Checking Homogeneity of Variance

### Residual vs Fitted Plot

```

library(ggplot2)

# Create diagnostic data
diag_data <- data.frame(
  fitted = fitted(model_crd),
  residuals = resid(model_crd),
  treatment = crd_data$treatment
)

ggplot(diag_data, aes(x = fitted, y = residuals)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(aes(color = treatment), size = 3, alpha = 0.7) +
  scale_color_manual(values = c("N0" = "#9D2235",
                                "N50" = "#FF8C42",
                                "N100" = "#2E8B57",
                                "N150" = "#4A90E2")) +
  labs(
    title = "Residuals vs Fitted Values",
    subtitle = "Random scatter = good; funnel shape = heteroscedasticity",
    x = "Fitted Values",
    y = "Residuals"
  ) +
  theme_minimal(base_size = 14)

```

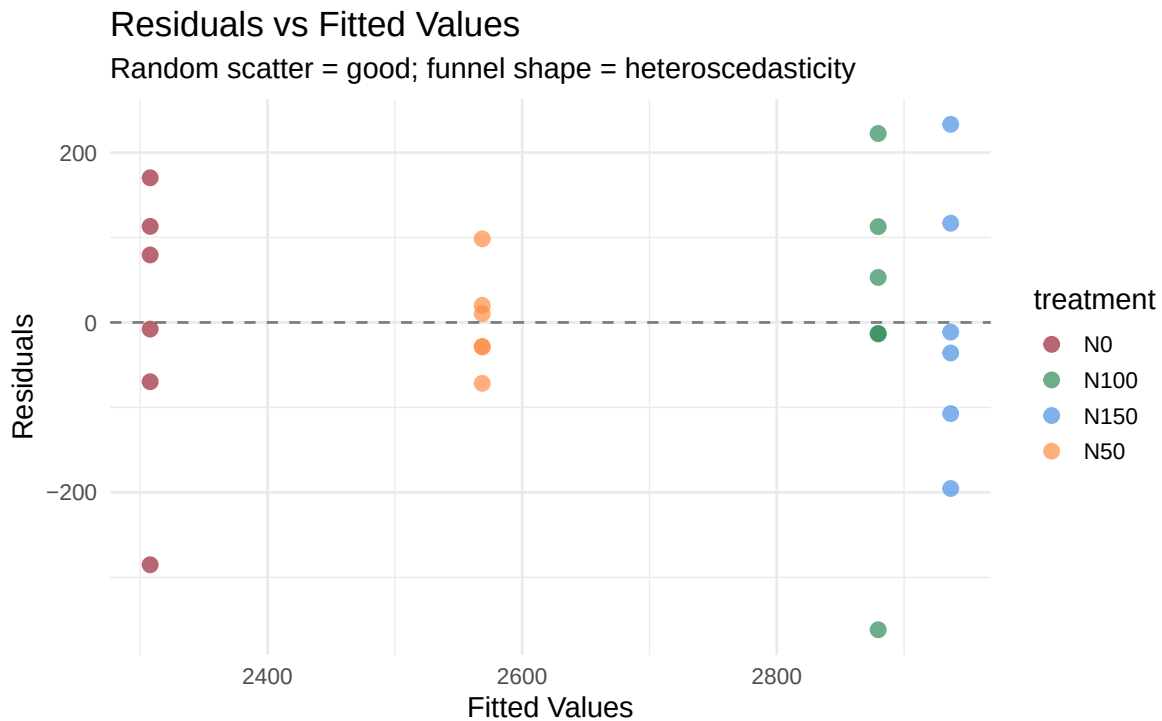


Figure 5: Residual vs fitted plot to check homoscedasticity

#### What to look for:

- **Good:** Random scatter around zero, no pattern
- **Bad:** Funnel shape (variance increases), systematic trends

**Timing:** ~3 minutes

**Teaching tip:** “If you see a megaphone or funnel, variance is not constant.”

#### Scale-Location Plot

```
diag_data <- diag_data |>
  dplyr::mutate(sqrt_abs_resid = sqrt(abs(residuals)))

ggplot(diag_data, aes(x = fitted, y = sqrt_abs_resid)) +
  geom_point(aes(color = treatment), size = 3, alpha = 0.7) +
  geom_smooth(method = "loess", se = FALSE, color = "#FF8C42", linewidth = 1) +
  scale_color_manual(values = c("N0" = "#9D2235",
```

```

    "N50" = "#FF8C42",
    "N100" = "#2E8B57",
    "N150" = "#4A90E2")) +
labs(
  title = "Scale-Location Plot",
  subtitle = "Flat smooth line = constant variance",
  x = "Fitted Values",
  y = " $\sqrt{|\text{Standardized Residuals}|}$ "
) +
theme_minimal(base_size = 14)

```

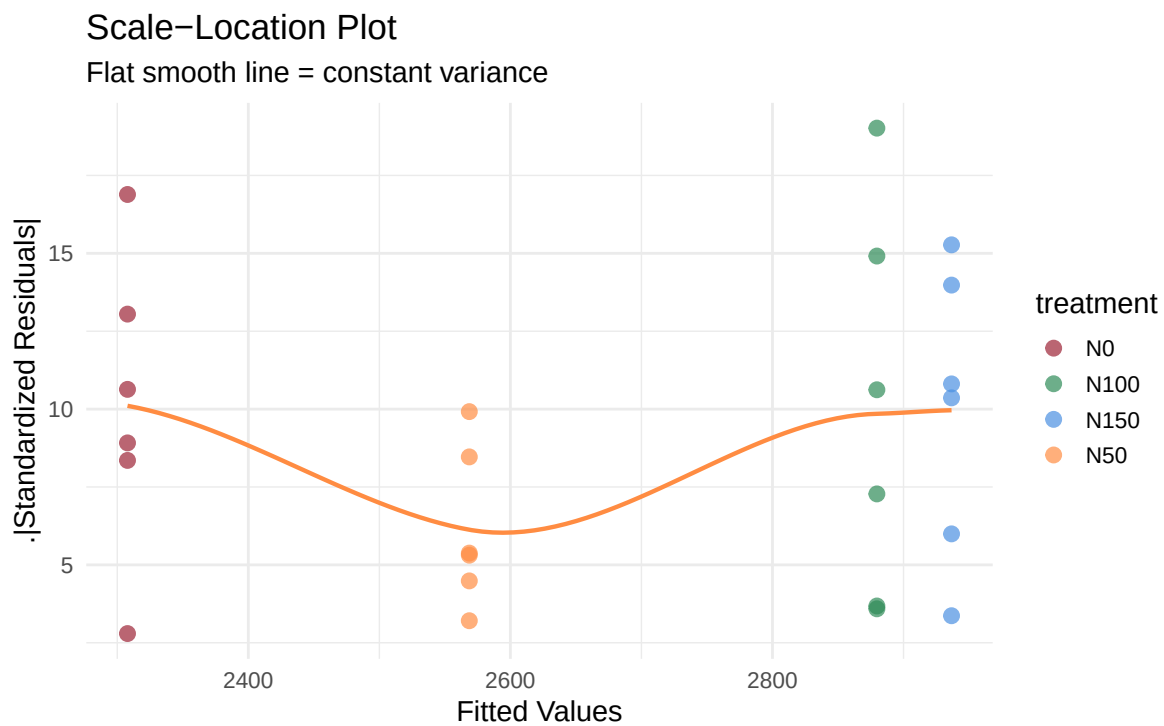


Figure 6: Scale-location plot: Checking for heteroscedasticity

#### Interpretation:

- Smooth line should be roughly horizontal
- Upward or downward trend → heteroscedasticity

**Timing:** ~3 minutes

**Explain:** “This plot amplifies variance patterns by plotting square root of absolute residuals.”

## Levene's Test for Equal Variance

**Formal hypothesis test:**

- $H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_t^2$  (Equal variances)
- $H_a$ : At least one variance differs

```
library(car)

# Levene's test
leveneTest(yield ~ treatment, data = crd_data)
```

```
#> Levene's Test for Homogeneity of Variance (center = median)
#>      Df F value Pr(>F)
#> group  3  0.9206 0.4489
#>      20
```

**Interpretation:**

- $p = 0.4489 > 0.05 \rightarrow$  Fail to reject  $H_0$
- No evidence of unequal variances
- Homoscedasticity assumption satisfied

### Caution

**Warning:** Like Shapiro-Wilk, Levene's test is sensitive to sample size. Use plots + test together.

**Timing:** ~3 minutes

**Decision rule:** If Levene's test  $p < 0.05$  AND plots show clear funnel, consider transformation.

# When Assumptions Fail: Transformations

## Common Data Transformations

### Transformation Table:

| Problem            | Transformation          |
|--------------------|-------------------------|
| Right skew         | $\log(y)$ or $\sqrt{y}$ |
| Count data         | $\sqrt{y + 0.5}$        |
| Proportion data    | $\arcsin(\sqrt{y})$     |
| Heteroscedasticity | $\log(y)$ often helps   |
| Don't know         | Box-Cox finds optimal   |

### When to transform:

- Q-Q plot shows clear skewness
- Residual plot shows funnel shape
- Variance ratio  $> 3:1$  (?) across groups
- Scientific precedent (e.g., log for enzyme activity)

**Remember:** Transform  $\rightarrow$  Analyze  $\rightarrow$  Interpret on original scale (back-transform means if needed)

**Timing:** ~4 minutes

**Reassurance:** “Transformation is normal in research, not a failure. Many biological processes are naturally log-scale.”

## Box-Cox Transformation

Finds optimal power transformation:

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(y) & \text{if } \lambda = 0 \end{cases}$$

```
library(MASS)

# Box-Cox on CRD model
bc <- boxcox(model_crd, lambda = seq(-2, 2, 0.1))
```

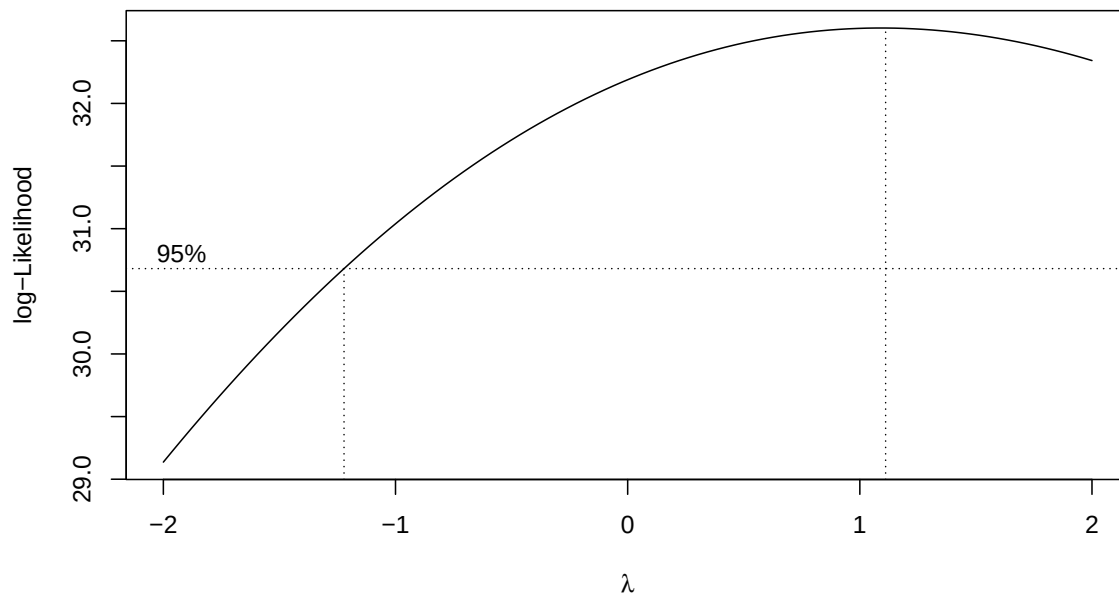


Figure 7: Box-Cox transformation plot: Optimal  $\lambda$  near 1 (no transformation needed)

```
# Extract optimal lambda
lambda_optimal <- bc$x[which.max(bc$y)]

cat("Optimal  $\lambda$  =", round(lambda_optimal, 2))
```

```
#> Optimal  $\lambda$  = 1.11
```

#### Interpretation:

- $\lambda = 1 \rightarrow$  No transformation
- $\lambda = 0.5 \rightarrow$  Square root
- $\lambda = 0 \rightarrow$  Log
- $\lambda = -1 \rightarrow$  Reciprocal

**Timing:** ~4 minutes

**For this example:**  $\lambda = 1.11$ , so original scale is fine (which we expected—data were simulated normally).

## Transformation Example: Right-Skewed Data

```
# Simulate right-skewed yield data (common in real trials)
set.seed(2026)
skewed_data <- data.frame(
  treatment = rep(c("A", "B", "C"), each = 10),
  yield = c(
    exp(rnorm(10, 5, 0.5)), # Treatment A
    exp(rnorm(10, 7, 0.5)), # Treatment B
    exp(rnorm(10, 9, 0.5))  # Treatment C
  )
)

# Fit model
model_skewed <- lm(yield ~ treatment, data = skewed_data)

# Check normality
shapiro.test(resid(model_skewed))
```

```
#>
#> Shapiro-Wilk normality test
#>
#> data:  resid(model_skewed)
#> W = 0.83621, p-value = 0.0003231
```

**Result:**  $p = 0.0003231 < 0.05 \rightarrow$  Significant non-normality

**Timing:** ~3 minutes

**Setup:** “This mimics real agricultural data where yield is often right-skewed.”

## Applying Log Transformation

```
# Log transformation
skewed_data$log_yield <- log(skewed_data$yield)

# Fit model on transformed data
model_log <- lm(log_yield ~ treatment, data = skewed_data)

# Check normality again
shapiro.test(resid(model_log))
```

```
#>
#> Shapiro-Wilk normality test
#>
#> data: resid(model_log)
#> W = 0.94608, p-value = 0.1326
```

**Result:**  $p = 0.1326 > 0.05 \rightarrow$  Normality assumption now satisfied!

**Re-run ANOVA:**

```
library(car)
#another option for ANOVA table
Anova(model_log)
```

```
#> Anova Table (Type II tests)
#>
#> Response: log_yield
#>           Sum Sq Df F value    Pr(>F)
#> treatment 104.720  2  202.49 < 2.2e-16 ***
#> Residuals   6.982 27
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Transformation fixed the violation.** Now interpret results on log scale or back-transform means for reporting.

**Timing:** ~4 minutes

**Key message:** “See how transformation can rescue a problematic analysis? This is a standard workflow.”

## Before vs After Transformation

**Timing:** ~2 minutes

**Visual comparison:** “Left shows clear deviation (bad). Right shows points on line (good).”

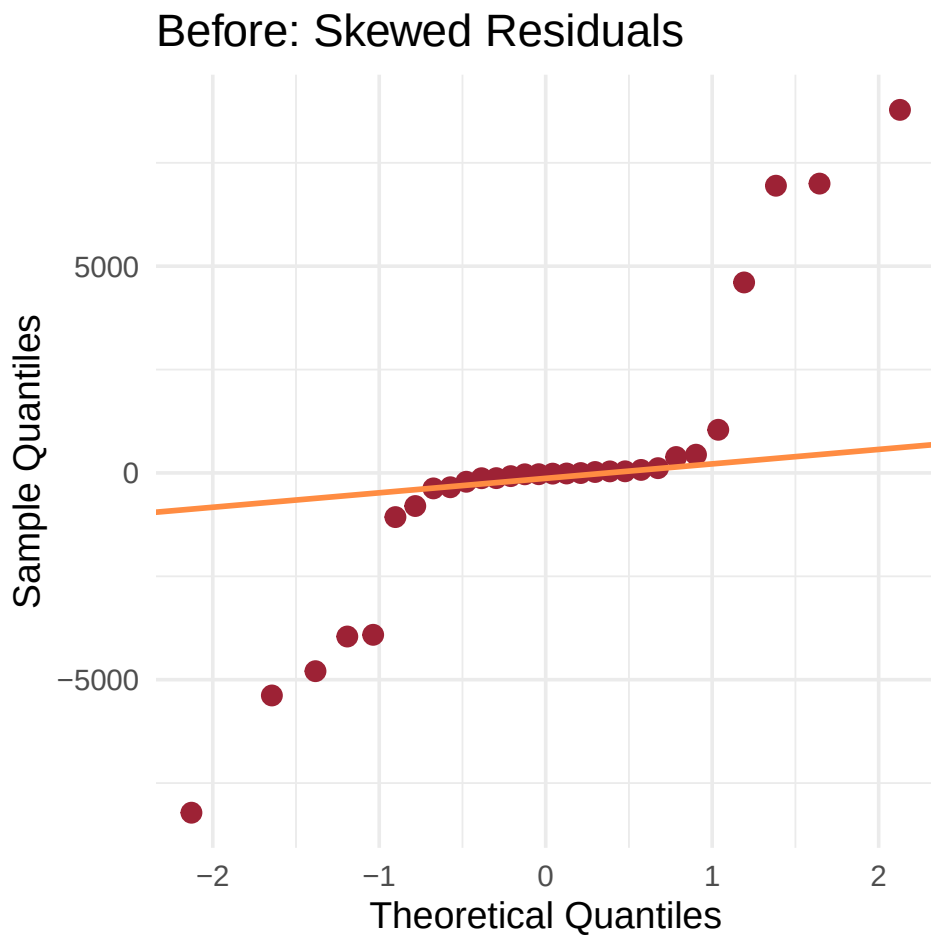


Figure 8: Q-Q plot BEFORE log transformation

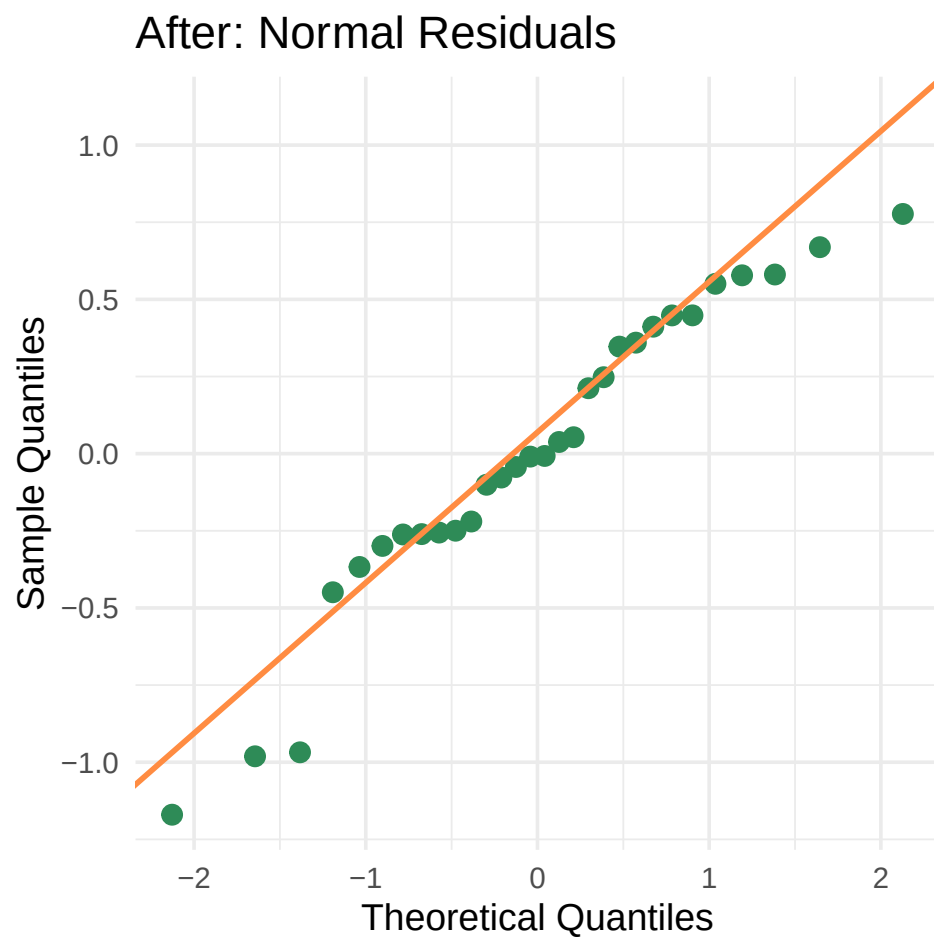


Figure 9: Q-Q plot AFTER log transformation

## Your Turn Activity

### Diagnose This Dataset

**Scenario:**

A food scientist tests 4 pasteurization temperatures on bacterial count (CFU/mL) in milk samples. CRD with 6 samples per temperature.

**Data provided:**

```
#>   temperature cfu
#> 1         60C 113
#> 2         60C 133
#> 3         60C 101
#> 4         60C 121
#> 5         60C 138
#> 6         60C 125
#> 7         65C  68
#> 8         65C  64
```

**Questions (3 minutes):**

1. Fit ANOVA model
2. Check Q-Q plot and residual plot
3. Is transformation needed? If so, which one?
4. Share your diagnosis with a neighbor

**Timing:** 3 minutes work + 2 minutes debrief

**Answer:** Count data likely needs sqrt transformation. Residuals probably right-skewed on original scale.

## Summary & Key Takeaways

- **Transformations** fix many violations: log, sqrt, Box-Cox

### What We Learned Today

- **Three ANOVA assumptions:** Normality, homogeneity of variance, independence
- **Diagnostic tools:** Q-Q plots, residual plots, Shapiro-Wilk, Levene's test
- **Visual diagnostics** are primary—formal tests are secondary
- **Normality violations** are less serious with balanced designs and large sample sizes

- **Heteroscedasticity** (unequal variances) is more serious—always address it

**Key principle:** Never trust ANOVA results without checking assumptions first!

**Timing:** ~3 minutes

## Resources

### Textbook:

- Oehlert (2010), **Chapter 3 and 6 (diagnostics)**: Model diagnostics and transformations

### R Packages:

- `car`: `leveneTest()`, `qqPlot()`, `Anova()`
- `MASS`: `boxcox()` for transformation selection
- `ggplot2`: Custom diagnostic plots

### Online Resources:

- Q-Q plot interpretation: [link](#)